ST. MARY'S DSG KLOOF

PHYSICAL SCIENCES PAPER 1 (Physics)

GRADE 10 **TIME: 2 HOURS + 10 min reading**
November 2025 **MARKS: 135**

Examiner: T. Kirsten **Moderators: K. Bolton and G. Main**

NAME: _____

Physical Sciences teacher: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of **22 pages**. Please check that your question paper is complete and write your name on it. A **data sheet** is also provided.

2. Read the questions carefully.

3. It is in your own interest to write legibly and to present your work neatly.

4. Answer ALL questions on the paper.

5. **All working must be shown** and units may be used in substitutions, but **MUST** be given with answers. Work to 2 decimal places or significant figures where appropriate.

6. Make sure your calculator is in the correct mode.

7. Extra space on page 22 is included at the end of the question paper.
   If you run out of space for a question, use this extra space. Clearly indicate the question number of your answer, should you use this extra space.

Question	1	2	3	4	5	6	7	Total
Your mark								
Mark	12	22	15	21	20	18	27	135

QUESTION 1 – MULTIPLE CHOICE

Answer the multiple-choice questions on the answer grid below. Make a clear cross (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

A	B	C	D

Here the option C has been marked as an example.

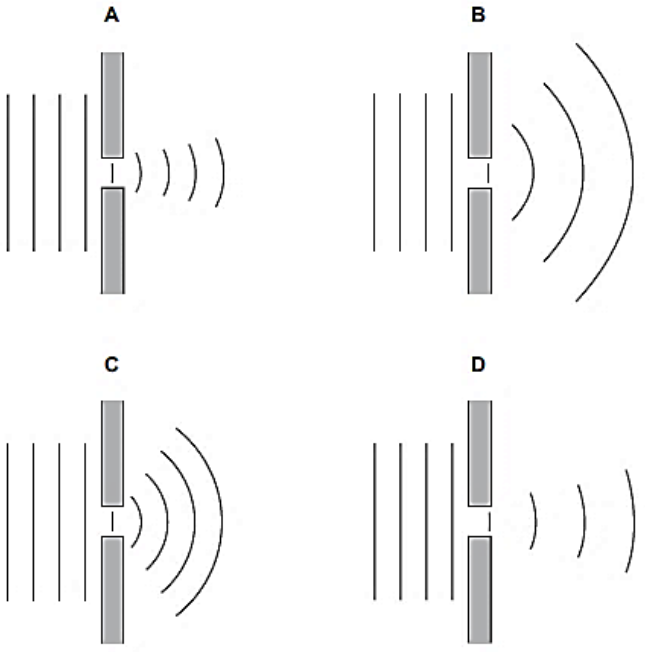
1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D

1.1 The frequency of a wave is doubled. The speed of the wave does not change. What happens to the wavelength of the wave?

A It halves
B It doubles
C It becomes four times as large
D It does not change

1.2 Plane water waves approach a narrow gap in a barrier.

Which diagram shows the most correct diffraction pattern that would occur?



1.3 According to the domain theory of magnetism, what happens inside a piece of iron when it becomes magnetised by an external magnetic field?

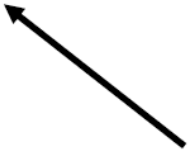
A The number of magnetic domains increases, and they become randomly aligned.

B The size of each magnetic domain decreases so that the atoms can spin faster.

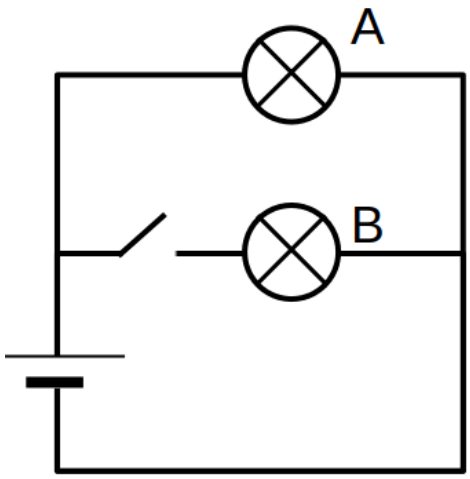
C The magnetic domains align more in the direction of the external field, increasing the net magnetic field of the material.

D The magnetic poles inside each domain switch places, creating new north and south poles at random positions.

1.4 Vector **x** and vector **y** are shown below. Which of the following is the correct resultant vector of vectors **x** and **y** given in the first column?

Vector x 	A. 	B. 
Vector y 	C. 	D. 

1.5 Two identical light bulbs, A and B, are connected to a cell as shown. What will happen to light bulb A when the switch is closed?



A Light bulb A will turn off.

B There will be no change to light bulb A.

C Light bulb A will brighten.

D Light bulb A will dim.

1.6 Which of the following statements about a cyclist moving on a circular track at a constant speed is **incorrect**?

A The cyclist's average velocity for one complete rotation is 0.

B Since the magnitude of the cyclist's speed is constant, the magnitude of his velocity is constant.

C The magnitude of the cyclist's instantaneous velocity equals his instantaneous speed.

D The cyclist experiences negative acceleration.

6 x 2 = [12]

QUESTION 2 WAVES

2.1 A bat in a cave emits a pulse of sound of wavelength 8,5 x 10^{-3} m. The speed of sound in the air of the cave is 340 $m.s^{-1}$. The pulse of sound hits the wall of the cave and is reflected back to the bat 0,12 s after it was emitted.



2.1.1 Define *frequency.* (2)

2.1.2 Calculate the frequency of the sound. Write your answer in scientific notation. (4)

2.1.3 Given the magnitude of the frequency calculated in Q2.1.2, can this type of sound be classified as infrasound, audible sound or ultrasound? (1)

2.1.4 Calculate the distance of the cave wall from the bat. (3)

2.1.5 Another bat is flying between the original bat and the wall of the cave. How would this other bat *hear* the sound of the pulse emitted by the first bat flying *towards* him, as compared to the actual sound emitted? (1)

2.1.6 Sketch a labelled diagram of the type of wave the bats emit. On your diagram label the parts of the wave, as well as one wavelength. Also indicate the direction the particles of the wave move. (4)

2.2 Three types of electromagnetic radiations are given in the table below.

Radiation
X-rays
Ultraviolet
Infra-red

2.2.1 Give one characteristic that all electromagnetic waves have in common. (1)

2.2.2 Write down the NAME of:

i) The radiation with the longest wavelength from the given list. (1)

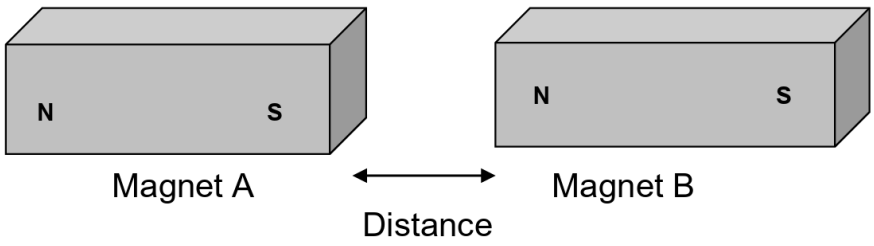
ii) The radiation whose photons have the greatest amount of energy. (1)

2.2.3 Calculate the energy of a photon of light which has a wavelength of 700 nm. (4)

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QUESTION 3 MAGNETISM

3.1 Refer to the following diagram of two identical magnets, a distance apart.



3.1.1 Define *magnetic field*. (2)

3.1.2 What would the direction of the magnetic field be between magnet A and magnet B?

Circle the correct answer: LEFT to RIGHT / RIGHT to LEFT (1)

3.1.3 Magnet A is cut into two smaller magnets. Indicate the poles on the smaller magnets. (2)



3.2 A student wanted to investigate if varying the distance between the two magnets affected the magnetic force between them.

- She did some research, and found that in previous experiments conducted, the mathematical relationship required the distance to be squared to obtain accurate results.
- With the help of her Physics teacher, she set up an apparatus which measures the magnetic force between magnets.
- She varied the distance between the magnets, and measured the force which corresponded to each distance.
- She did a calculation to square the distance between the magnets.

Her results were as follows:

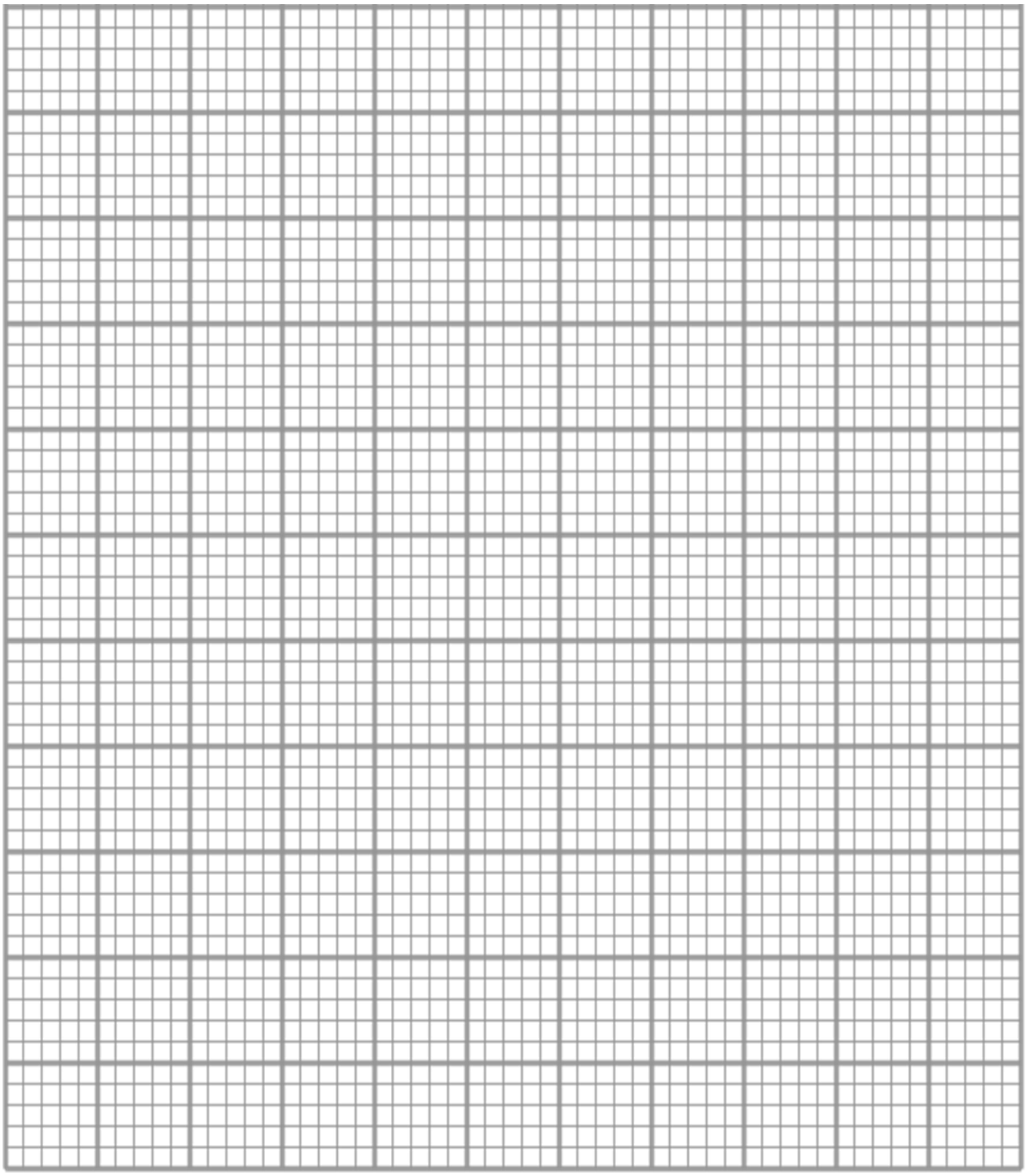
Distance between magnets squared (mm^2)	**Magnetic force (N)**
50	2
120	1,6
260	1,1
640	0,6
950	0,4

3.2.1 Which was the dependent variable in this investigation? (1)

3.2.2 List one variable which would have to be controlled in this investigation. (1)

3.2.3 On the graph paper provided alongside, plot a graph of the student's results. (5)

Graph to show the relationship between the distance between magnets squared and the magnetic force between them.



3.2.4 State the student's conclusion, using a scientific relationship. (3)

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QUESTION 4 ELECTROSTATICS

4.1 A balloon is rubbed on a child's hair in Diagram A and then held in position in Diagram B. A negatively charged rod is brought close to the balloon, and the rod *repels* the balloon.



Diagram A



Diagram B

4.1.1 What charge does the balloon have after being rubbed on the child's hair? (1)

4.1.2 Explain how the balloon came to have the charge stated in Q4.1.1. (2)

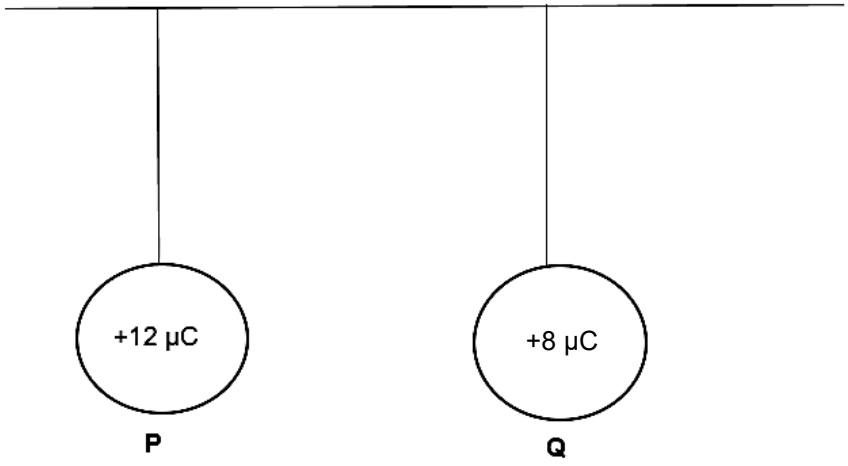
4.1.3 If 300 electrons were transferred between the balloon and the hair, what is the new charge on the balloon and hair respectively? (4)

4.2 Every charge in the universe consists of integer multiples of the electron charge.

4.2.1 What is this principle called? (1)

4.2.2 A charged object carries a charge of −8 nC. How many excess electrons are present on the object? (3)

4.3 Two identical metal spheres, P and Q, are suspended on light, inelastic cotton threads. They carry charges of +12 µC and +8 µC, respectively.



4.3.1 Draw the electric field pattern between the two charges. (3)

4.3.2 The two spheres make contact and electrons are transferred from one sphere to the other. Will the transfer of electrons be from P to Q or Q to P? (1)

The spheres are now separated after making contact.

4.3.3 Calculate the new charge on the spheres. (3)

4.3.4 How many electrons are transferred? (3)

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QUESTION 5 KINEMATICS

5.1 Some Grade 10 girls set out on cycle ride on The Journey.

- The girls initially cycle for 3 km South along a straight road.
- They then turn West, and cycle for a further 1,7 km before stopping at a waterfall.



5.1.1 Define *displacement*. (2)

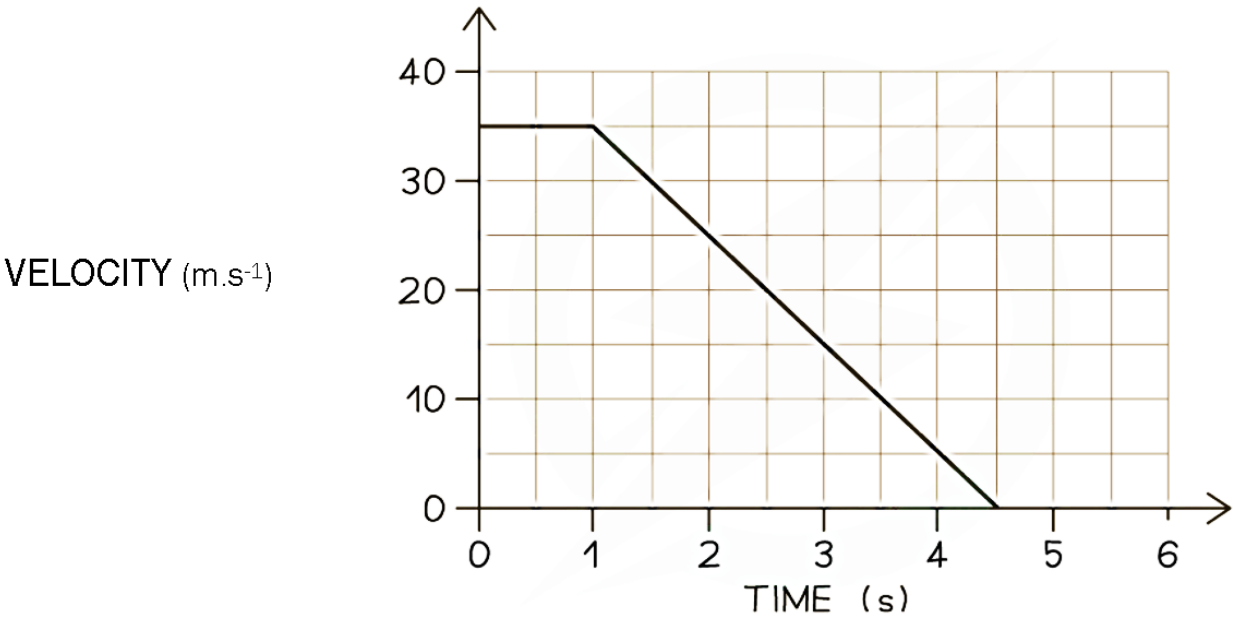
5.1.2 Calculate the resultant displacement of the cyclists from the starting point to the waterfall. (4)

From their position 1,7 km West, the girls then turn and cycle back to the starting point, following the same route.

5.1.3 Define *distance*. (2)

5.1.4 If the entire cycling time to see the waterfall and return to the starting point took 50 minutes, calculate the average speed of the cyclists during the ride. Give your answer in $m.s^{-1}$. (4)

5.2 While driving along a road in an Easterly direction, Kate sees a dog in the road ahead of her at time t = 0 s. The velocity-time graph below shows how the speed of the car changes as Kate reacts and then presses the brake pedal, bringing the car to a halt.



5.2.1 What was Kate's reaction time? (1)

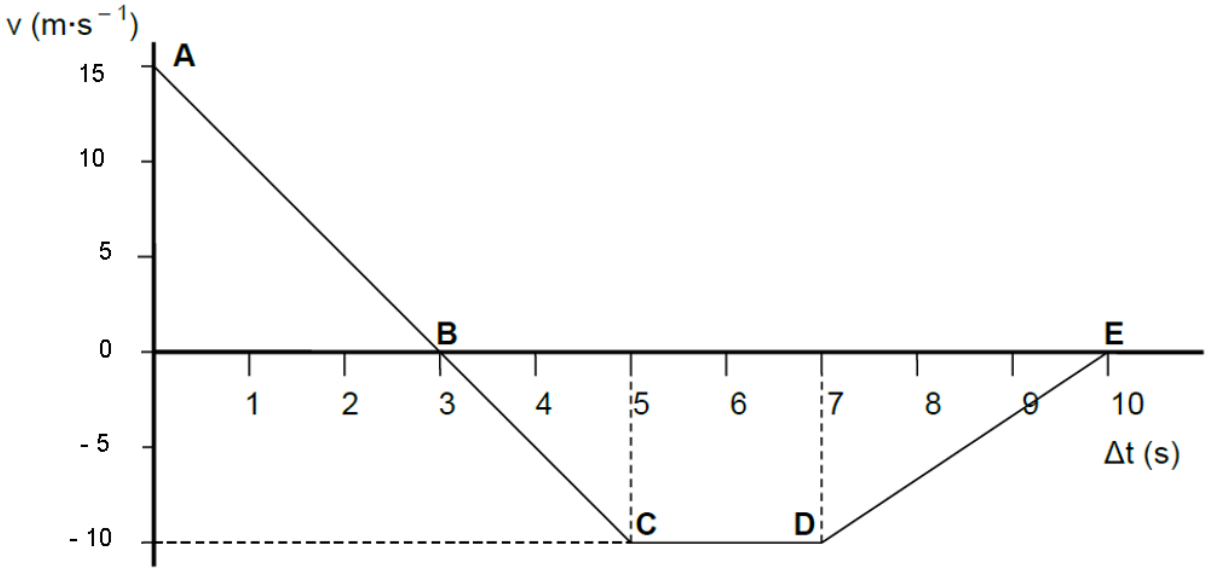
5.2.2 Determine the braking distance of the car. (3)

5.2.3 Calculate the acceleration of the car during the braking distance. (4)

[20]

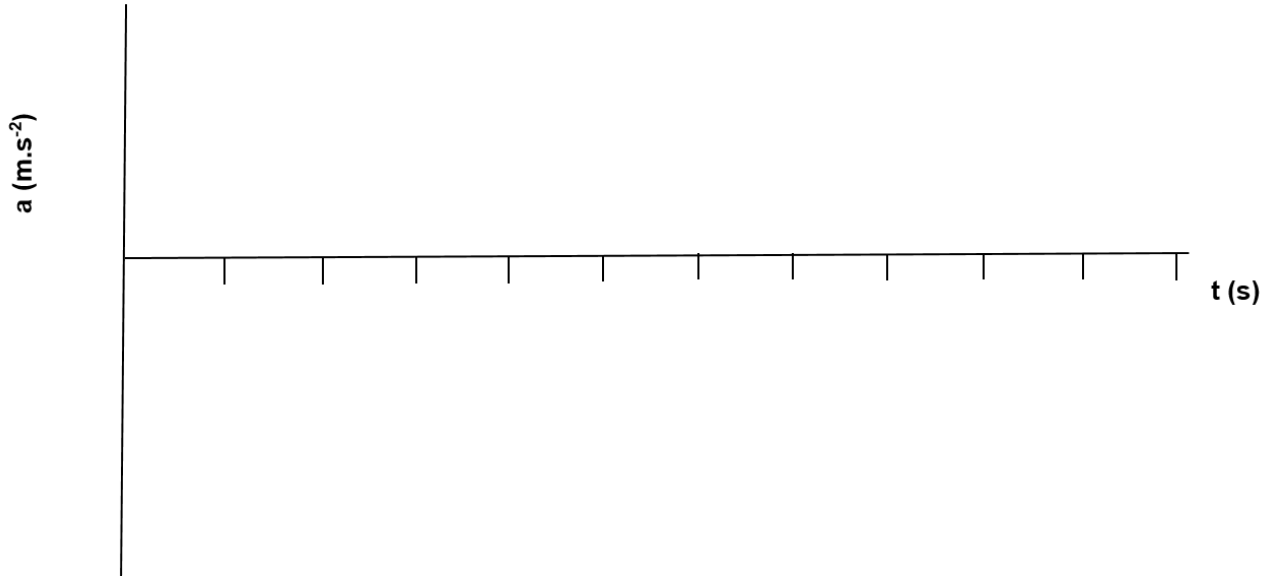
QUESTION 6

6.1 The graph below shows the changing velocity of an object over a 10 s period. The object was initially moving North.

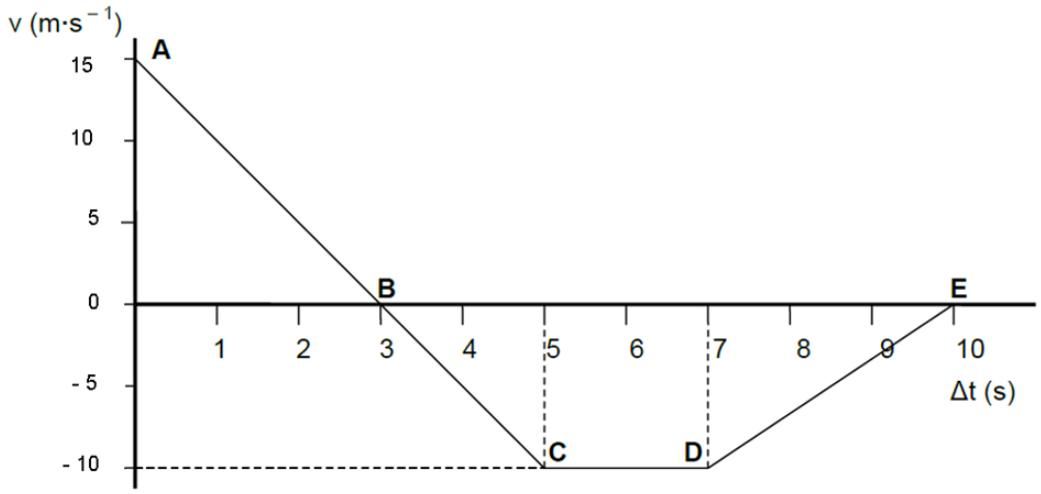


6.1.1 Describe the motion of the object from D to E, in terms of velocity and acceleration. (4)

6.1.2 **Sketch** an acceleration time graph for the object over 10 seconds on the axes below. Only the time values and the *relative* magnitudes of the acceleration values need be shown. (4)



6.1.3 The total displacement of the object over the 10 s time interval is 22,5 m South. Draw a position time sketch graph to show the motion of the object over the 10 second time interval. Clearly indicate the intervals AB, BC, CD and DE for each section of the motion. No displacement values need be shown, it is the shape which is important.
(The velocity time graph is reproduced again below for ease of reference.) (6)



6.2 A truck is travelling at a constant speed of 7,8 $m.s^{-1}$ West when it passes a stationary car at the side of the road.

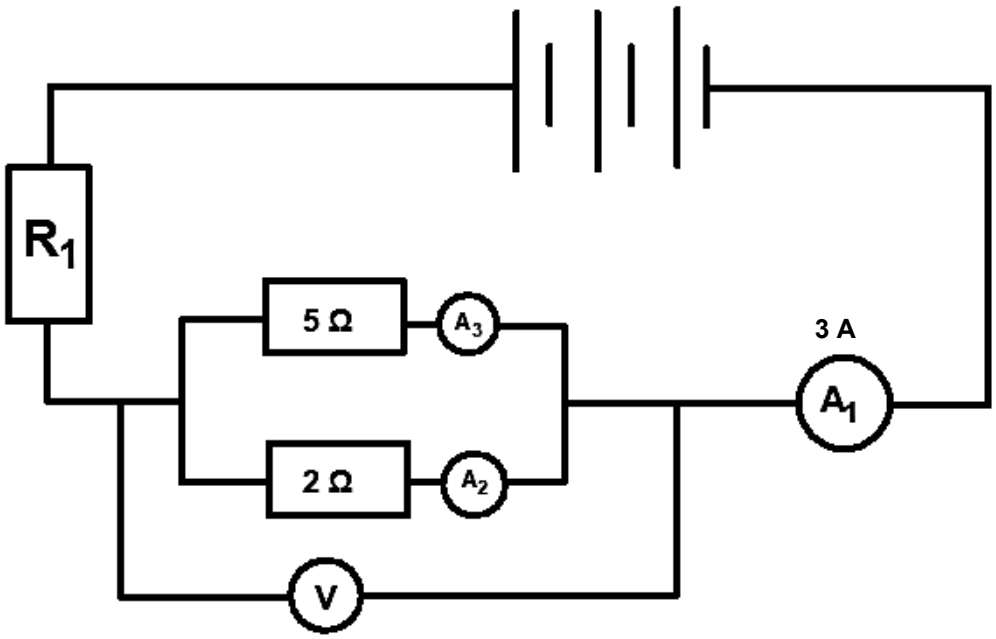
- When the truck is a distance of **x** metres ahead of the stationary car, the car accelerates uniformly from rest at 4,2 $m.s^{-2}$, West.
- It takes the car 8 s to catch up to the truck.

Calculate the original distance **x** between the car and the truck. (4)

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QUESTION 7 ELECTRIC CIRCUITS

7.1 In the circuit diagram below, **each cell is 3 V**, and A_1 reads 3 A.



7.1.1 Calculate the total resistance of the circuit. (3)

7.1.2 Calculate the effective resistance of the parallel combination. (2)

7.1.3 Calculate the resistance of R_1. (2)

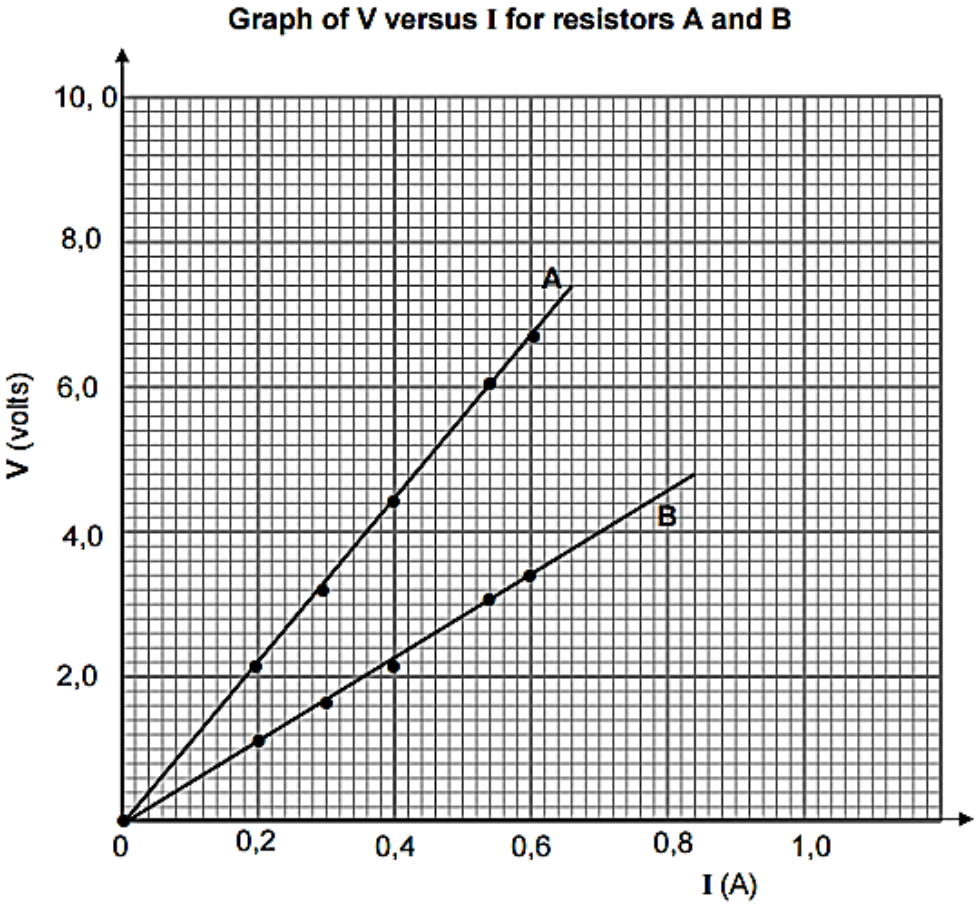
7.1.4 Calculate the reading on the voltmeter. (2)

7.1.5 Calculate the reading on A_2. (3)

7.1.6 How much work is done in pushing 5 C of charge through the 2 Ω resistor? (3)

7.2 The cost of electricity is R3.20 per kWh. A 3 000 W electrical oven is used for 45 minutes. Calculate the cost of using the oven for 45 minutes. (4)

7.3 Two Grade 10 pupils want to construct an electric heater using one of two wires, A or B, made of different materials. They conduct experiments and draw the graphs as shown below.



7.3.1 Apart from temperature, write down TWO other factors that the learners should consider to ensure a fair test when choosing which wire to use. (2)

7.3.2 Using the graph, calculate the resistance of Wire A and Wire B. (5)

Wire A	**Wire B**

7.4 Assuming all other factors are kept constant, state which ONE of the two wires, A or B, will be the most suitable to use in the heater. (1)

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EXTRA SPACE